

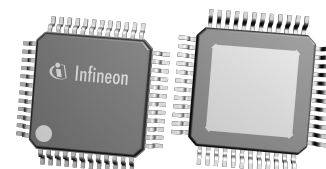
TLE9015DQU

iso UART transceiver IC



Features

- Compatibility for ring mode topology
- Two UART ports for serial communication to host microcontroller
- Two independent iso UART interfaces for communication to other BMS ICs
- 3 Mbit/s data rate for fast communication
- Fully transparent communication scheme from UART to iso UART
- Integrated internal logic for minimizing pin count on the UART side
- Differential current edge triggered iso UART communication interface
- High robustness against external noise
- Two external fault inputs (EMM and ERRQ_ext)
- Latching error output pin for triggering external microcontroller
- Internal supply monitoring
- Green Product (RoHS compliant)



Potential applications

Hybrid electric vehicles (HEV), plug-in hybrid electric vehicles (PHEV), battery electric vehicles (BEV) as well as in 12 V/48 V Li-ion batteries as well as energy storage systems (ESS).

Product validation

Qualified for automotive applications. Product validation according to AEC-Q100.

Description

The device is a general purpose transceiver IC to be used in battery systems for enabling the communication between the main host microcontroller and the cell supervision ICs which are usually connected to the battery module potential. The IC is designed for Li-Ion battery packs used in hybrid electric vehicles (HEV), plug-in hybrid electric vehicles (PHEV), battery electric vehicles (BEV) as well as stationary Li-Ion batteries.

Additionally to the physical layer translation, the device offers the possibility to communicate potential detected errors in a cell inside the battery pack to the main microcontroller.

Type	Package	Marking
TLE9015DQU	PG-TQFP-48	TLE9015DQU

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1 Block diagram

1 Block diagram

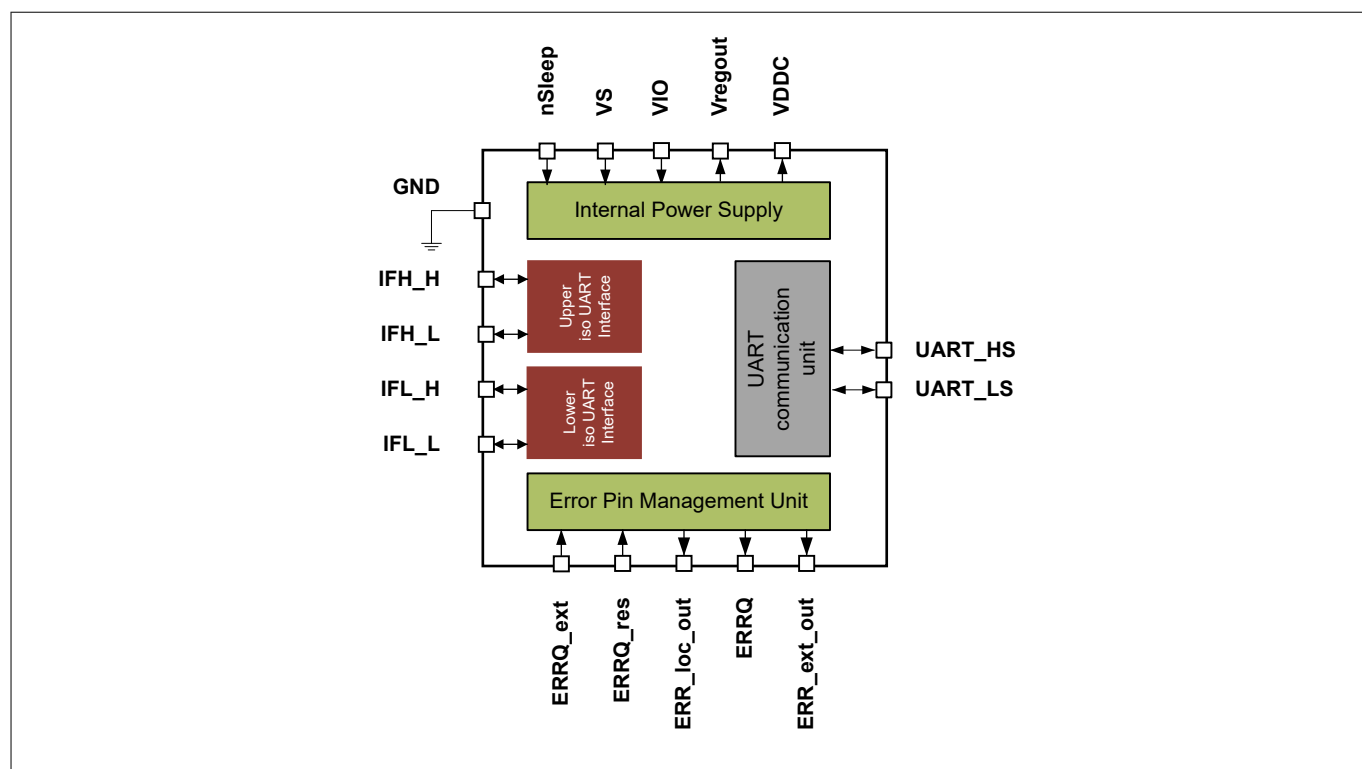


Figure 1 Block diagram

2 Pin configuration

2 Pin configuration

2.1 Pin assignment

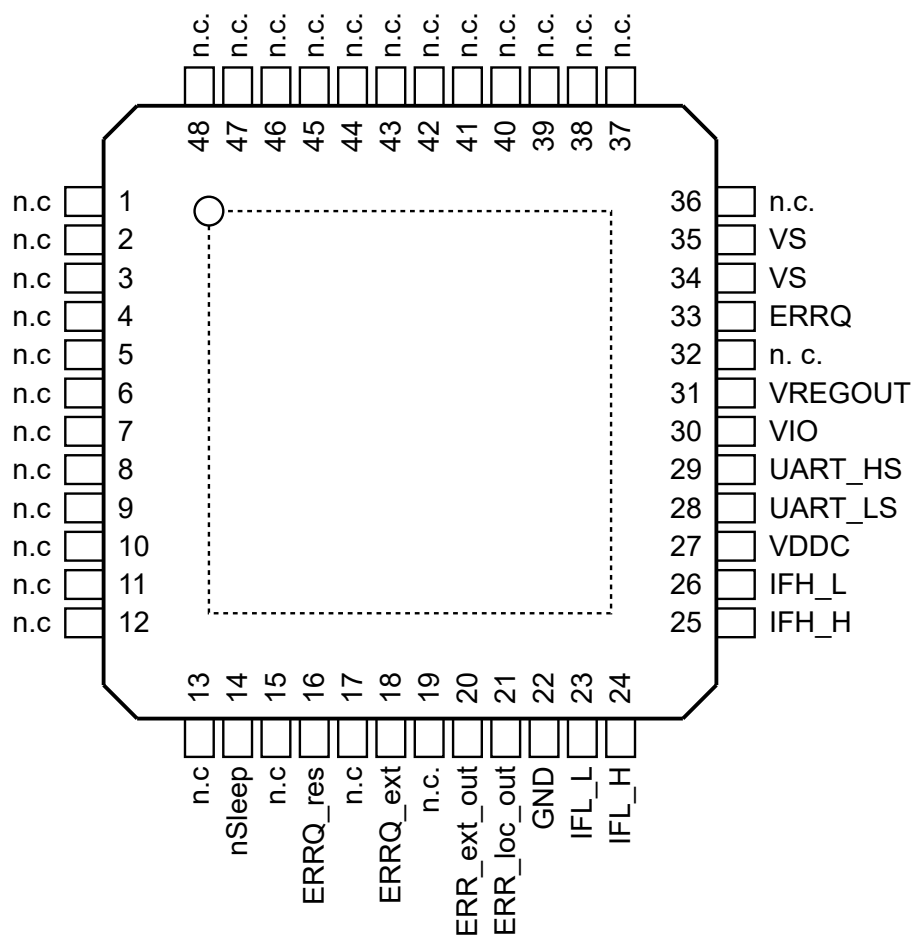


Figure 2 Pin configuration (top view)

2.2 Pin definitions and functions

Pin	Symbol	Pin type	Function
1	n. c.		Not connected. Connect to GND in application.
2	n. c.		Not connected. Connect to GND in application.

2 Pin configuration

Pin	Symbol	Pin type	Function
3	n. c.		Not connected. Connect to GND in application.
4	n. c.		Not connected. Connect to GND in application.
5	n. c.		Not connected. Connect to GND in application.
6	n. c.		Not connected. Connect to GND in application.
7	n. c.		Not connected. Connect to GND in application.
8	n. c.		Not connected. Connect to GND in application.
9	n. c.		Not connected. Connect to GND in application.
10	n. c.		Not connected. Connect to GND in application.
11	n. c.		Not connected. Connect to GND in application.
12	n. c.		Not connected. Connect to GND in application.
13	n. c.		Not connected. Connect to GND in application.
14	nSleep	I	Input pin to force the device to go to sleep; active low.
15	n. c.		Not connected. Connect to GND in application.
16	ERRQ_res	I	Reset the ERRQ pin; active "low".
17	n. c.		Not connected. Connect to GND in application.
18	ERRQ_ext	I	Input for external ERRQ. If not used, connect to VREGOUT.
19	n. c.		Not connected. Connect to GND in application.
20	ERR_ext_out	O	Output for external ERRQ; active "high".
21	ERR_loc_out	O	Output pin to indicate local ERRQ; active "high".
22	GND	GND	Local GND of the device.
23	IFL_L	D_I / O	Lower isolated UART (iso UART) L pin.
24	IFL_H	D_I / O	Lower isolated UART (iso UART) H pin.
25	IFH_H	D_I / O	Upper isolated UART (iso UART) H pin.
26	IFH_L	D_I / O	Upper isolated UART (iso UART) L pin.
27	VDDC	Supply	Buffer capacitor pin for internal iso UART supply.
28	UART_LS	D_I / O	UART LS channel.
29	UART_HS	D_I / O	UART HS channel.
30	VIO	S	Supply for UART interface.
31	VREGOUT	S	Output pin for the internal regulator.
32	n.c.	n.c.	Not connected. Connect to GND in application.
33	ERRQ	HV_D_O	Error pin; open drain NMOS. This pin is latching.
34	VS	S	Supply pin.
35	VS	S	Supply pin.
36	n. c.		Not connected. Connect to GND in application.

2 Pin configuration

Pin	Symbol	Pin type	Function
37	n. c.		Not connected. Connect to GND in application.
38	n. c.		Not connected. Connect to GND in application.
39	n. c.		Not connected. Connect to GND in application.
40	n. c.		Not connected. Connect to GND in application.
41	n. c.		Not connected. Connect to GND in application.
42	n. c.		Not connected. Connect to GND in application.
43	n. c.		Not connected. Connect to GND in application.
44	n. c.		Not connected. Connect to GND in application.
45	n. c.		Not connected. Connect to GND in application.
46	n. c.		Not connected. Connect to GND in application.
47	n. c.		Not connected. Connect to GND in application.
48	n. c.		Not connected. Connect to GND in application.
–	Exposed Pad	GNDA	Cooling tab. Connect to GND in the application.

Pin types: A = analog, D = digital, HV = high-voltage, I = input, O = output, I/O = bidirectional, P = power, S = supply

3 General product characteristics

3 General product characteristics

Within the functional or operating range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given in the electrical characteristics table.

This thermal data was generated in accordance with JEDEC JESD51 standards. For more information, go to www.jedec.org.

3.1 Absolute maximum ratings

Table 1 Absolute maximum ratings

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
Voltages							
Supply voltage VS	V _{VS_max}	-0.3	–	45	V	–	PRQ-1368
Supply voltage VS relative	V _{VS_rel_max}	V _{VREGOUT} - 0.3	–	–	V	–	PRQ-489
Supply voltage VIO	V _{VIO_max}	-0.3	–	5.5	V	–	PRQ-488
Regulator output VREGOUT	V _{VREGOUT_max}	-0.3	–	3.6	V	–	PRQ-490
Regulator output VDDC	V _{VDDC_max}	-0.3	–	3.6	V	Assuming I _{VDDC} ≤ 1 mA continuous current	PRQ-491
iso UART interface IFL_x	V _{IFL_L_max} V _{IFL_H_max}	-4.1	–	6.6	V	¹⁾ BCI test maximum 300 mA injected via twisted pair cable onto iso UART interface (maximum pin current 150 mA)	PRQ-493
iso UART interface IFH_x	V _{IFH_L_max} V _{IFH_H_max}	-4.1	–	6.6	V	¹⁾ BCI test maximum 300 mA injected via twisted pair cable onto iso UART interface (maximum pin current 150 mA)	PRQ-492
Ground pin GND	V _{GND}	0	–	0	V	Absolute GND	PRQ-511
High voltage input pin nSleep	V _{nSleep_max}	-0.3	–	45	V	²⁾	PRQ-524
High voltage input pin ERRQ_res	V _{ERRQ_res_max}	-0.3	–	45	V	²⁾	PRQ-526

(table continues...)

3 General product characteristics

Table 1 (continued) **Absolute maximum ratings**

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
High voltage input pin ERRQ_ext	$V_{\text{ERRQ_ext_max}}$	-0.3	–	45	V	²⁾	PRQ-528
Digital output pin absolute ERR_ext_out	$V_{\text{ERR_ext_out_t_max}}$	-0.3	–	5.5	V	²⁾	PRQ-531
Digital output pin relative ERR_ext_out	$V_{\text{ERR_ext_out_t_rel_max}}$	-0.3	–	$V_{\text{VIO}} + 0.3$	V	²⁾	PRQ-530
Digital output pin absolute ERR_loc_out	$V_{\text{ERR_loc_out_t_max}}$	-0.3	–	5.5	V	²⁾	PRQ-533
Digital output pin relative ERR_loc_out	$V_{\text{ERR_loc_out_t_rel_max}}$	-0.3	–	$V_{\text{VIO}} + 0.3$	V	²⁾	PRQ-532
UART interface pins absolute	$V_{\text{UART_x_ma}_x}$	-0.3	–	5.5	V	²⁾ $x \rightarrow \text{LS or HS}$	PRQ-535
UART interface pins relative	$V_{\text{UART_x_rel_max}}$	-0.3	–	$V_{\text{VIO}} + 0.3$	V	²⁾ $x \rightarrow \text{LS or HS}$	PRQ-534

ESD robustness

ESD robustness 2 kV	$V_{\text{ESD_2kV_max}}$	-2	–	2	kV	³⁾ HBM; all pins	PRQ-514
ESD robustness 4 kV	$V_{\text{ESD_4kV_max}}$	-4	–	4	kV	³⁾ HBM; robustness versus GND for pins: VS, IFH_x, IFL_x	PRQ-1831
ESD robustness CDM 500 V	$V_{\text{ESD_cdm_all_max}}$	-500	–	500	V	⁴⁾ CDM; all pins	PRQ-516
ESD robustness CDM 750 V	$V_{\text{ESD_Corner_max}}$	-750	–	750	V	⁴⁾ CDM; corner pins	PRQ-517

(table continues...)

3 General product characteristics

Table 1 (continued) Absolute maximum ratings

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			

Temperatures

Junction temperature	T_{j_max}	-40	–	150	$^{\circ}\text{C}$	–	PRQ-512
Storage temperature	T_{stg_max}	-55	–	150	$^{\circ}\text{C}$	–	PRQ-513

- 1) Positive and negative transients with a maximum duration of 100 ns allowed between $\pm 8\text{ V}$; This should simulate ESD events; however, during normal and steady-state condition voltage on these pins must stay inside the maximum ratings specified.
- 2) Not subject to production test, specified by design.
- 3) ESD robustness, HBM according to ANSI/ESDA/JEDEC JS-001 (1.5 k Ω , 100 pF).
- 4) ESD robustness, Charged Device Model JESD22-C101.

Notes:

1. Stresses above the ones listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.
2. Integrated protection functions are designed to prevent IC destruction under fault conditions described in the datasheet. Fault conditions are considered as outside normal operating range. Protection functions are not designed for continuous repetitive operation.

3.2 Functional range

Table 2 Functional range

$T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
Supply voltage VS	$V_{VS_functional}$	4.75	–	45	V	–	PRQ-1367
Supply voltage VIO	$V_{VIO_functional}$	3	–	5.5	V	–	PRQ-520

3 General product characteristics

3.3 Thermal resistance

Table 3 Thermal resistance

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
Junction to case	R_{thJC}	–	20	–	K/W	1)	PRQ-1846
Junction to ambient	R_{thJA}	–	48	–	K/W	1) 2)	PRQ-1847

1) Not subject to production test, specified by design.

2) Specified R_{thJA} value is according to JEDEC JESD51-5,-7 at natural convection on FR4 2s2p board; The product (chip and package) was simulated on a $76.2 \times 114.3 \times 1.5$ mm board with 2 inner copper layers ($2 \times 70 \mu\text{m Cu}$, $2 \times 35 \mu\text{m Cu}$). The thermal via array under the exposed pad consists of 16 vias with a diameter of 0.3 mm and a plating thickness of 25 μm .

3.4 Definitions of voltages and currents

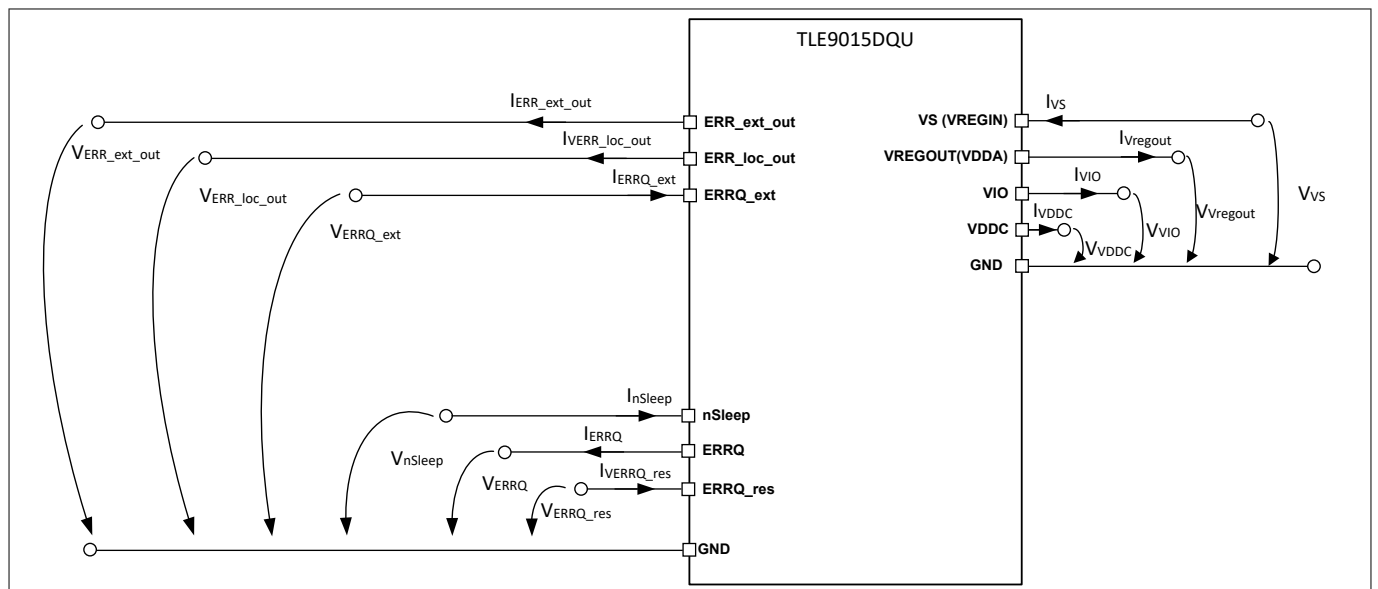


Figure 3 Definitions of voltages and currents

4 Power Management Unit (PMU)

4 Power Management Unit (PMU)

4.1 Functional description

The transceiver IC can be powered from an external LDO via VS pin or any other source which can supply the voltage V_{VS} .

To supply the communication interface, the device provides a regulated output voltage V_{VDDC} on pin VDDC.

The device provides a regulated output voltage $V_{VREGOUT}$ with an output current $I_{VREGOUT}$ on pin VREGOUT which can supply the GPIOs of the device or other loads.

The voltage at the VIO pin sets the logic levels and supplies the GPIOs. The pin can be connected directly to the VREGOUT pin or to another desired voltage level using an external regulator.

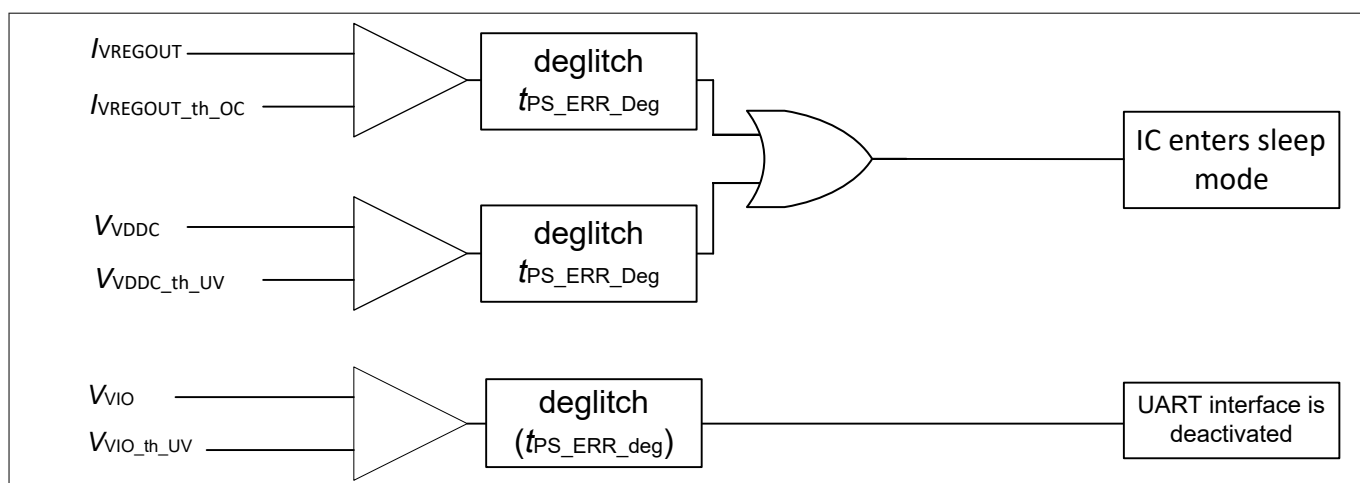


Figure 4 Power supply monitoring

The IC is forced to go to sleep mode via the nSleep pin. The pin is edge triggered from "high" to "low" and has an internal pull-up resistor R_{nSleep_PU} .

4.2 Electrical characteristics power management unit (PMU)

Table 4 Electrical characteristics

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
Internal regulators							
VREGOUT internal regulator output voltage	V _{VREGOUT}	3.3	3.45	3.6	V	–	PRQ-544
VDDC output voltage	V _{VDDC}	2.42	2.5	2.63	V	–	PRQ-549

(table continues...)

4 Power Management Unit (PMU)

Table 4 (continued) Electrical characteristics

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
Supply currents							
VS sleep mode current	I_{VS_sleep}	–	3	23	μA	1. typical value $T_j = 25^{\circ}\text{C}$ 2. $-40^{\circ}\text{C} < T_j < 50^{\circ}\text{C}$	PRQ-1341
VS idle current	I_{VS_idle}	–	4.9	5.5	mA	IC in idle mode	PRQ-557
VREGOUT current consumption multi purpose supply	$I_{VREGOUT}$	–	–	10	mA	No load on VIO	PRQ-1373
VIO current consumption during UART communication	I_{VIO_comm}	–	–	5	mA	No load on VREGOUT	PRQ-695
VS current consumption during communication	I_{VS_comm}	–	$I_{VS_idle_typ} + 0.9$	$I_{VS_idle_max} + 1.2$	mA	- UART communication. - Current to charge external interface components not included.	PRQ-694
VS current consumption during iso UART communication including external interface components	$I_{VS_comm_isoU}$	–	–	$I_{VS_comm} + 7.6$	mA	¹⁾ 1. $C_{ser} = 1\text{ nF}$ 2. $BR_{iso_U} = 2\text{ Mbit/s}$ 3. $R_{ser} = 39\ \Omega$ 4. $C_{isoUART_F} = 220\text{ pF}$ 5. Valid for one iso UART interface in TX mode	PRQ-562

Protection and Detection

VREGOUT overcurrent threshold	$I_{VREGOUT_th_OC}$	31	40	60	mA	Tested during idle mode	PRQ-545
VIO undervoltage threshold falling	$V_{VIO_th_UV_fall}$	2.2	–	2.76	V	–	PRQ-546

(table continues...)

4 Power Management Unit (PMU)

Table 4 (continued) **Electrical characteristics**

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
VIO undervoltage threshold rising	$V_{VIO_th_UV_rise}$	2.24	–	2.9	V	–	PRQ-547
VIO undervoltage threshold hysteresis	$V_{VIO_th_UV_hys}$	40	100	160	mV	–	PRQ-548
VDDC undervoltage threshold	$V_{VDDC_th_Uv}$	2.15	–	2.42	V	–	PRQ-550
VDDC undervoltage threshold hysteresis	$V_{VDDC_th_Uv_hys}$	80	100	140	mV	–	PRQ-551
Power supply error detection deglitch time	$t_{PS_ERR_deg}$	8	15	24	μs	¹⁾	PRQ-552

nSleep pin

nSleep input range voltage "low"	V_{nSleep_LOW}	0	–	0.99	V	–	PRQ-690
nSleep input range voltage "high"	$V_{nSleep_HIG_H}$	2.52	–	V_{VS}	V	–	PRQ-691
nSleep internal pull up resistor	R_{nSleep_PU}	200	300	400	$k\Omega$	connected to Vregout	PRQ-692
nSleep input deglitch	$t_{nSleep_degltch}$	19.85	21.82	24	μs	¹⁾	PRQ-693

¹⁾ Not subject to production test; verified by design or characterization.

5 Watchdog and wake-up function (WD)

5 Watchdog and wake-up function (WD)

5.1 Functional description

The IC generates the wake-up pattern on:

- IFL, if the IC received a valid wake-up pattern on interface UART_HS.
 - (1) indicates the source of wake-up, (2) indicates the propagation on IFL_x
 - Ring mode: (5) indicates propagated wake-up signal received on IFH_x and (6) forwarded to UART_LS.
- IFH, if the IC received a valid wake-up pattern on interface UART_LS.
 - (3) indicates the source of wake-up, (4) indicates the propagation on IFH_x

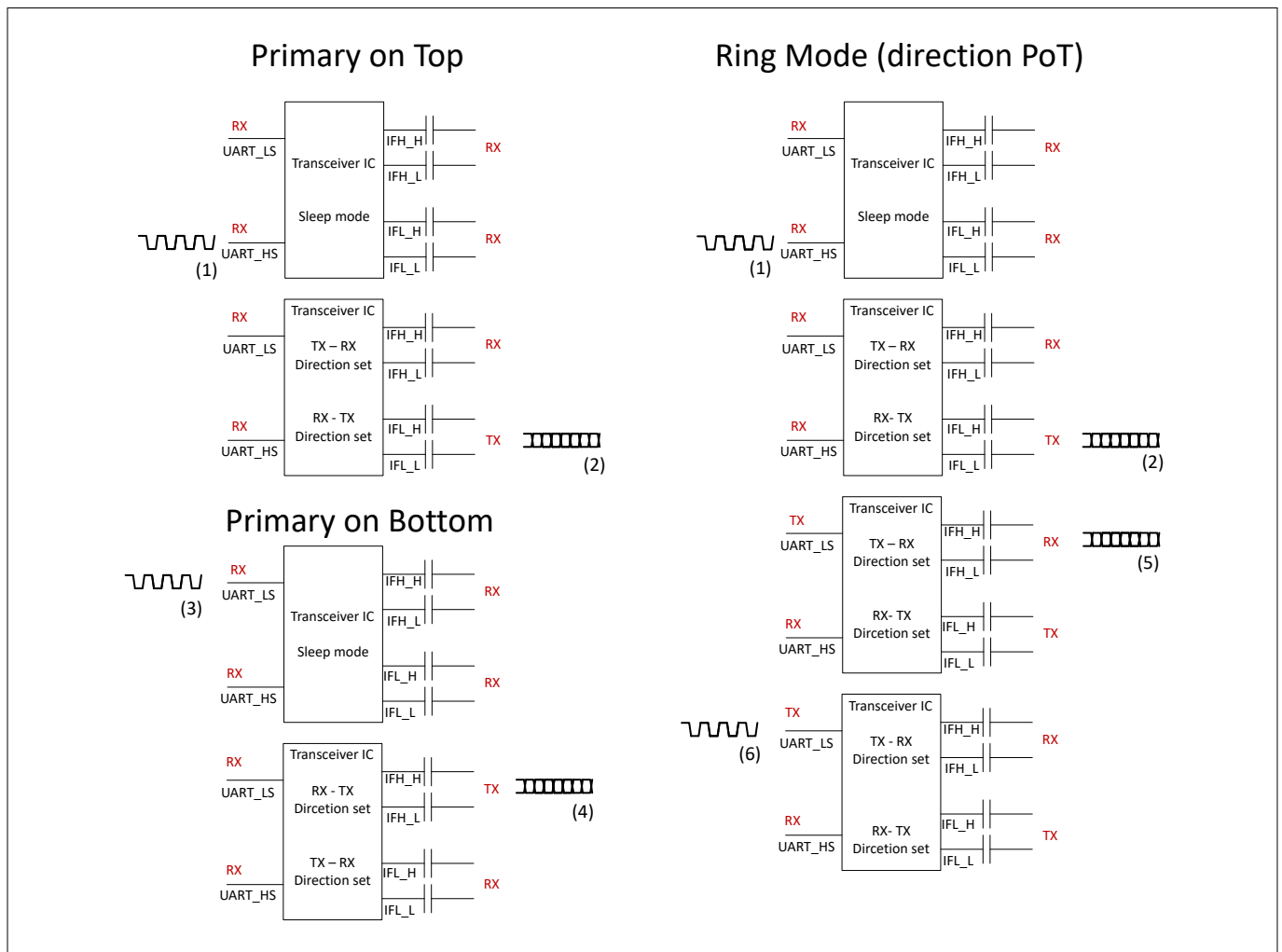


Figure 5 Wake-up signal propagation

Note: If EMM is received in sleep mode, the UART interfaces are configured in Tx mode, hence the device has to send into sleep mode first to reconfigure at least one UART interface to Rx.

5 Watchdog and wake-up function (WD)

5.2 Electrical characteristics watchdog and wake-up function (WD)

Table 5 **Electrical characteristics**

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
Wake-up function							
WD wake-up signal frequency	f_{WAKEUP}	48	50	1500	kHz	–	PRQ-572
WD device wake-up time	t_{WAKE}	200	370	500	μs	48 kHz wake-up frequency. From the first falling edge of the input pattern to the first edge of the propagated wake-up sequence.	PRQ-573
WD wake-up - number of detected periods	$n_{\text{WAKE_det}}$	4	–	8	periods	–	PRQ-574
WD wake-up propagation - length in periods	n_{WAKE}	8	–	8	periods	–	PRQ-575

6 Communication

6.1 Functional description

The device supports the following communication interfaces.

1. UART
2. iso UART

The IC acts as link between UART and iso UART interfaces supporting all communication modes:

- Ring mode
- Primary on bottom (PoB)
- Primary on top (PoT)
- Simultaneous PoB and PoT e.g in case of broken wire between sensing ICs iso UART interfaces

The device forwards a received message to the next device in the system. The time between receiving and forwarding the message, either on UART or isoUART is defined as $t_{\text{UART_isoU_del}}$

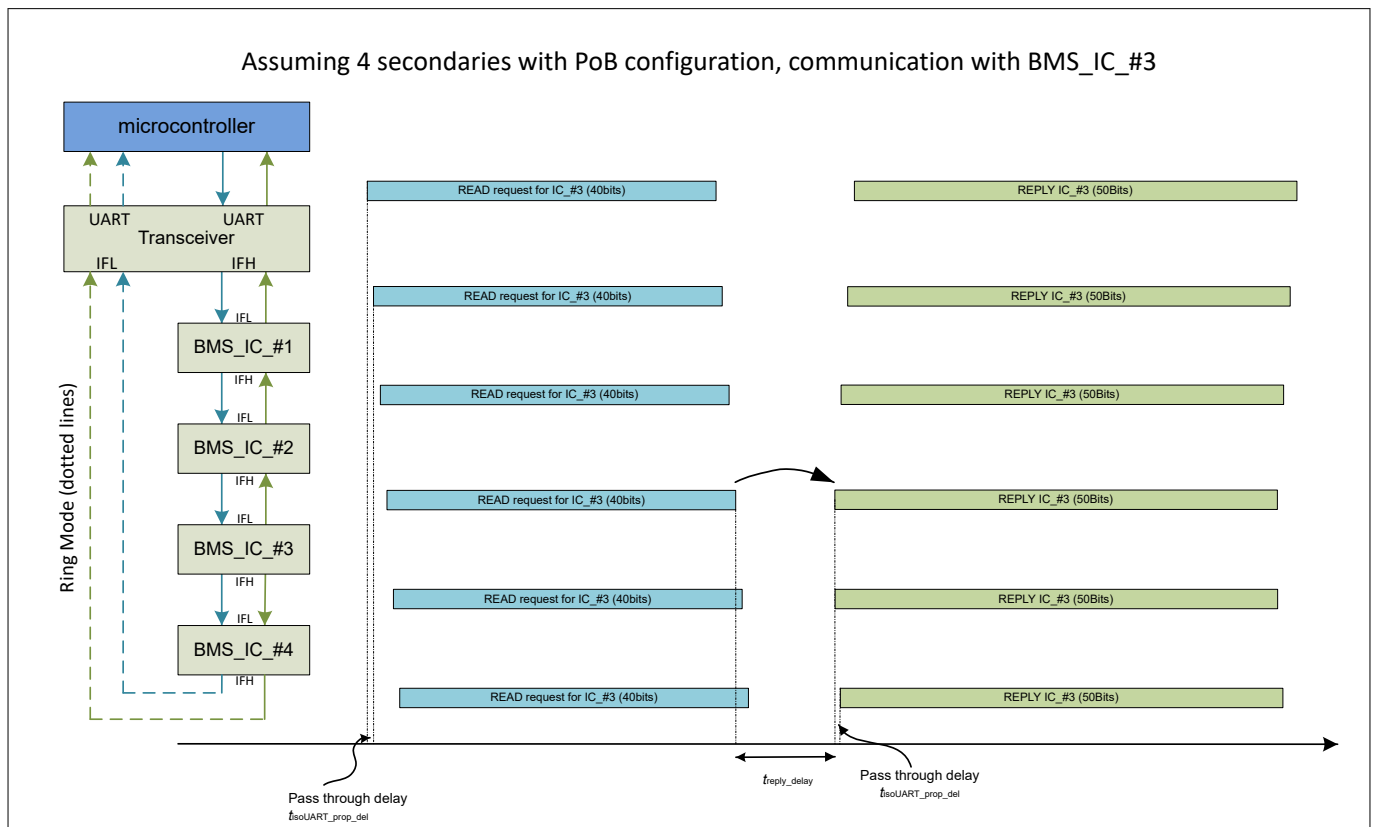


Figure 6 Communication propagation delays
iso UART waveform specification

6 Communication

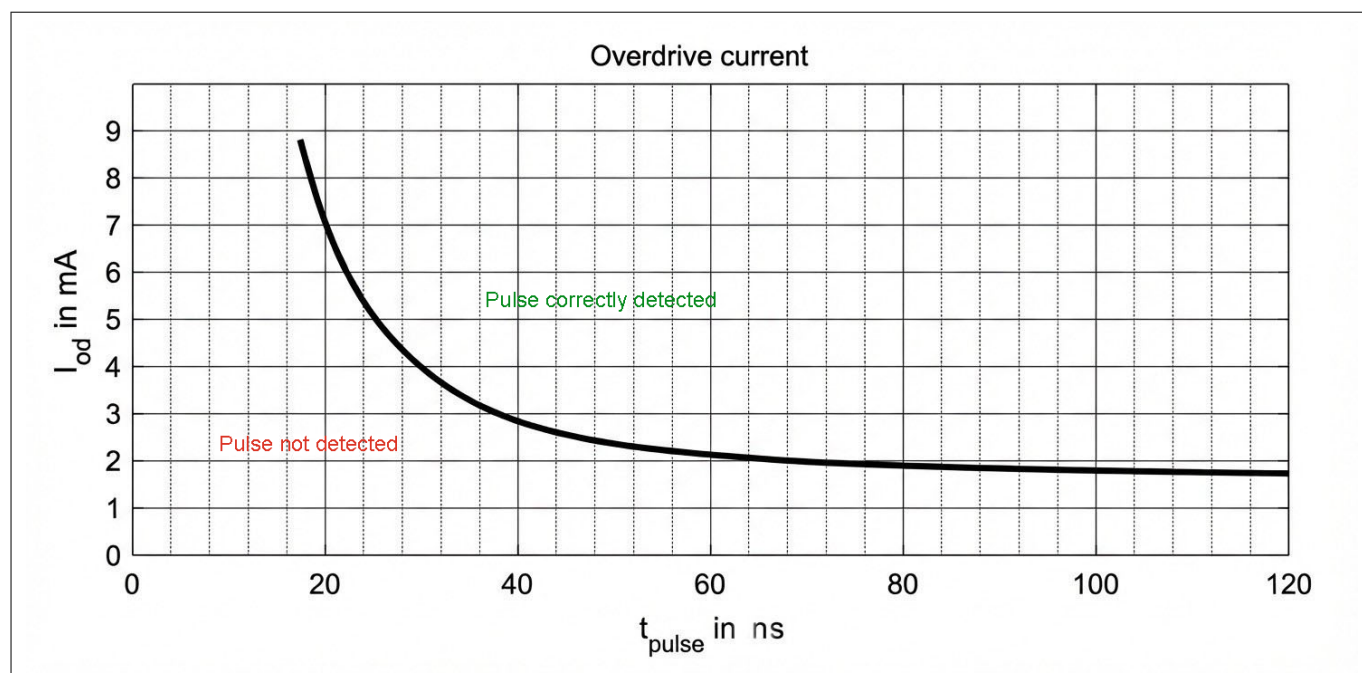


Figure 7 iso UART waveform specification

6.2 Electrical characteristics communication

Table 6 Electrical characteristics

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
UART physical layer							
UART input "low" level	$V_{\text{UART_low}}$	0	–	$V_{\text{VIO}} \times 0.3$	V	–	PRQ-719
UART input "high" level	$V_{\text{UART_high}}$	$V_{\text{VIO}} \times 0.7$	–	V_{VIO}	V	–	PRQ-720
UART output "low" level	$V_{\text{UART_low}}$	0	–	0.45	V	$I_{\text{UART}} \leq 5 \text{ mA}$	PRQ-721
UART output high level	$V_{\text{UART_HIGH}}$	$V_{\text{VIO}} - 0.45$	–	V_{VIO}	V	$I_{\text{UART}} \geq -5 \text{ mA}$	PRQ-722
UART output current	I_{UART}	-5	–	5	mA	Current capability of UART output; x → LS or HS	PRQ-723
UART bit rate	BR_{UART}	0.97	2.0	3	Mbit/s	Determined by microcontroller. Device's responses are synchronized to microcontroller UART bit rate.	PRQ-724
External capacitance on UART pin	$C_{\text{UART_ext}}$	–	–	30	pF	–	PRQ-725

(table continues...)

6 Communication

Table 6 (continued) Electrical characteristics

$V_{\text{VS}} = V_{\text{VS_functional}}$, $T_j = -40^\circ\text{C}$ to $+150^\circ\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
UART to iso UART propagation delay	$t_{\text{UART_isoU_del}}$	–	25	60	ns	Propagation delay from UART to iso UART	PRQ-828
UART broadcast read bus release time	$t_{\text{UART_rel_BR}}$	–	–	$15 \times 1/BR_{\text{UA RT}}$	s	Time to wait before sending a new command after end of broadcast read reply	PRQ-1911
UART read, write, broadcast write, multiread release time	$t_{\text{UART_rel}}$	–	–	$3 \times 1/BR_{\text{UA RT}}$	s	Time to wait before sending a new command after end of read/write/broadcast write/multiread reply	PRQ-1912
iso UART current threshold "high"	$I_{\text{isoU_th_high}}$	2.25	4.5	6.5	mA	$(I_{\text{IFX_H}} - I_{\text{IFX_L}}) / 2$ $I_{\text{IFX_H}}$: Current in the iso UART high pin $I_{\text{IFX_L}}$: Current in the iso UART low pin	PRQ-832
iso UART current threshold "low"	$I_{\text{isoU_th_low}}$	-6.5	-4.5	-2.25	mA	$(I_{\text{IFX_H}} - I_{\text{IFX_L}}) / 2$ $I_{\text{IFX_H}}$: Current in the iso UART high pin $I_{\text{IFX_L}}$: Current in the iso UART low pin	PRQ-833
iso UART propagation delay	$t_{\text{isoU_prop_del}}$	–	25	70	ns	¹⁾ Propagation delay from IFH to UART_LS and IFL to UART_HS	PRQ-834
iso UART overdrive current	I_{od}	3	–	–	mA	²⁾ with $t_{\text{pulse}} = 38 \text{ ns}$	PRQ-1370
Reply delay time	$t_{\text{reply_delay}}$	0	1.7	3	μs	²⁾ internal reply delay time of one IC	PRQ-837
iso UART bit rate	BR_{isoU}	0.97	2	3	Mbit/s	–	PRQ-838
Series resistor value	R_{ser}	37.0 5	39	40.9 5	Ω	²⁾ ³⁾	PRQ-836
Series capacitor value	C_{ser}	0.95	1	1.05	nF	²⁾ ³⁾	PRQ-835
Transceiver Ron @100mA	R_{ON}	19	22	27	Ω	–	PRQ-1845

6 Communication

- 1) Tested with standard external circuit (C_{ser} , R_{ser}).
 - 2) Not subject to production test; verified by design or characterization.
 - 3) External RC network needs to be adjusted depending on the application constraints, for example cable length.
-

7 Emergency mode (EMM) and ERR pin (ERR)

7 Emergency mode (EMM) and ERR pin (ERR)

7.1 Functional description

The following events trigger the ERRQ pin:

- Incoming emergency mode signal (EMM) via iso UART interface
- Fault input pin ERRQ_ext

The emergency signal is an alternating signal with the frequency f_{EMM} . In case of an incoming EMM signal, the IC enters EMM and starts the fault handling.

The EMM signal has a duration of n_{EMM} periods. The number of periods the IC needs to detect an EMM signal is depending on the operation mode:

1. Idle mode: $n_{\text{EMM_dect}}$
2. Straight after wake-up caused by EMM: $n_{\text{EMM_dect_wake-up}}$

The output of the ERRQ pin is open drain. The "low" state is latching and can be reset by the ERRQ_res pin.

The ERR_loc_out pin indicates, whether an incoming EMM signal triggered the error handling procedure. On detection of an EMM signal, the device sets the ERR_loc_out pin "high" for the duration of $t_{\text{ERR_loc_out}}$. The logic levels of the pin are VIO and GND.

The ERR_ext_out pin is an output pin. It indicates that the fault was triggered by the ERRQ_ext pin and represents the deglitched and inverted ERRQ_ext signal. The pin is active high and the logic levels are VIO and GND.

The ERRQ_ext pin is an input pin to trigger the ERRQ pin externally. The pin is level triggered active low with a deglitch time of $t_{\text{ERRQ_ext_deglitch}}$. The pin has an internal pull-up resistor $R_{\text{ERRQ_ext_PU}}$ to the internal 3.3 V supply implemented.

The ERRQ_res pin is the input pin to reset the ERRQ pin function. The pin is level triggered active low. The pin has an internal pull-up resistor $R_{\text{ERRQ_res_PU}}$ to the internal 3.3 V supply implemented and includes a deglitch filter with deglitch time defined by $t_{\text{ERRQ_res_deglitch}}$.

The following table outlines the Error pin timings and response conditions:

Table 7 Behavior of the error interface pins

Type	Pin name	Pin level = LOW	Pin level = HIGH	Timing
Out	ERR_loc_out	No EMM	EMM detected	For $t_{\text{ERR_loc_out}}$ independent of EMM
Out	ERR_ext_out	ERR_ext_out = HIGH	ERR_ext_out = LOW	Until $t_{\text{ERR_ext_out}}$ deglitch elapsed
In	ERRQ_res	ERRQ = HIGH	–	Until $t_{\text{ERR_res}}$ deglitch elapsed
Out	ERRQ	EMM received	–	Until resetted e.g by ERRQ = HIGH
Out	ERRQ	ERRQ_ext = LOW	–	–
In	ERRQ_ext	ERRQ = LOW	–	–

7 Emergency mode (EMM) and ERR pin (ERR)

7.2 Electrical characteristics emergency mode (EMM) and ERR pin (ERR)

Table 8 Electrical characteristics

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
EMM number of periods to detect EMM signal - straight after wake-up	$n_{EMM_dect_wake-up}$	4	–	4	periods	¹⁾ IC just entered IDLE state straight after wake-up procedure.	PRQ-739
EMM number of periods to detect EMM signal - idle mode	n_{EMM_dect}	16	–	16	periods	¹⁾ IC is in idle mode	PRQ-741

ERRQ pins

ERRQ functional voltage range	V_{ERRQ}	4.75	–	V_{VS}	V	$V_{VS} \leq 20\text{ V}$	PRQ-718
ERRQ activated output voltage	V_{ERRQ_low}	0	–	0.3	V	$I_{ERRQ} \leq 1.5\text{ mA}$	PRQ-717
ERRQ sink current	I_{ERRQ}	–	–	1.5	mA	¹⁾	PRQ-1340
ERRQ_res input voltage range "low"	$V_{ERRQ_res_low}$	0	–	0.99	V	–	PRQ-704
ERRQ_res input voltage range "high"	$V_{ERRQ_res_high}$	2.52	–	V_{VS}	V	–	PRQ-705
ERRQ_ext input voltage range "low"	$V_{ERRQ_ext_low}$	0	–	0.99	V	–	PRQ-706
ERRQ_ext input voltage range "high"	$V_{ERRQ_ext_high}$	2.52	–	V_{VS}	V	–	PRQ-707
ERR_loc_out output voltage	$V_{ERR_loc_out_low}$	0	–	0.45	V	$I_{ERR_loc_out} \leq 5\text{ mA}$	PRQ-708

(table continues...)

7 Emergency mode (EMM) and ERR pin (ERR)

Table 8 (continued) Electrical characteristics

$V_{VS} = V_{VS_functional}$, $T_j = -40^{\circ}\text{C}$ to $+150^{\circ}\text{C}$, all voltages with respect to GND, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note or condition	P-Number
		Min.	Typ.	Max.			
ERR_loc_out output voltage "high"	$V_{ERR_loc_out_high}$	$V_{VIO} - 0.45$	–	V_{VIO}	V	$I_{ERR_loc_out} \geq -5 \text{ mA}$	PRQ-709
ERR_ext_out output voltage "low"	$V_{ERR_ext_out_low}$	0	–	0.45	V	$I_{ERR_ext_out} \leq 5 \text{ mA}$	PRQ-710
ERR_ext_out output voltage high	$V_{ERR_ext_out_high}$	$V_{VIO} - 0.45$	–	V_{VIO}	V	$I_{ERR_ext_out} \geq -5 \text{ mA}$	PRQ-711
ERRQ_ext internal pull-up resistor	$R_{ERRQ_ext_pu}$	200	300	400	k Ω	Connected to an internal 3.3 V supply	PRQ-712
ERRQ_res internal pull-up resistor	$R_{ERRQ_res_pu}$	200	300	400	k Ω	Connected to an internal 3.3 V supply	PRQ-713
ERRQ_ext input deglitch	$t_{ERRQ_ext_deglitch}$	47.2 6	49.7 4	52.4 2	ms	¹⁾	PRQ-714
ERRQ_res input deglitch	$t_{ERRQ_res_deglitch}$	844	951	1067	μs	¹⁾	PRQ-715
ERR_loc_out active time	$t_{ERR_loc_out}$	281. 32	293. 74	307. 2	ms	¹⁾	PRQ-716

¹⁾ Not subject to production test; verified by design or characterization.

8 Application information

8.1 External circuitry and components

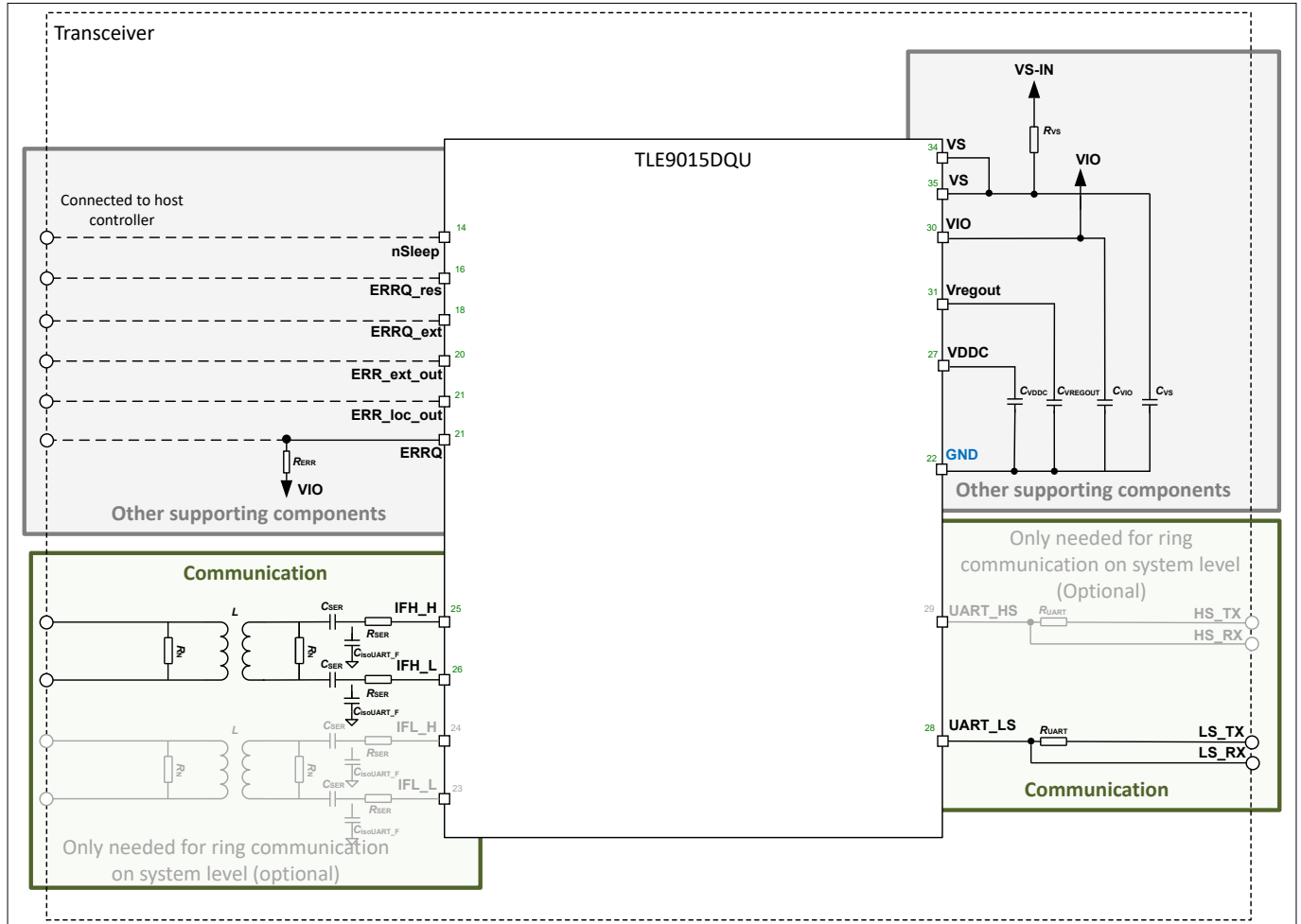


Figure 8

Table 9 External components

Buffer capacitor VS	C_{VS}	100	nF	–
Filtering resistor VS	R_{VS}	5.1	Ω	–
Buffer capacitor on VREGOUT	$C_{VREGOUT}$	100	nF	–
Buffer capacitor on VIO	C_{VIO}	100	nF	If VIO is connected to VREGOUT, then C_{VIO} is omitted.
Buffer capacitor on VDDC	C_{VDDC}	330	nF	–
Bypass capacitor on iso UART	$C_{isoUART_F}$	220	pF	–
Damping resistor	R_N	1	k Ω	With Pulse HM2116ANL transformer
UART network	R_{UART}	1	k Ω	$\tau \leq 50$ ns

(table continues...)

8 Application information

Table 9 **(continued) External components**

Transformer	L	–	–	Transformer L from Pulse HM2116ANL.
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9 Package information

9 Package information

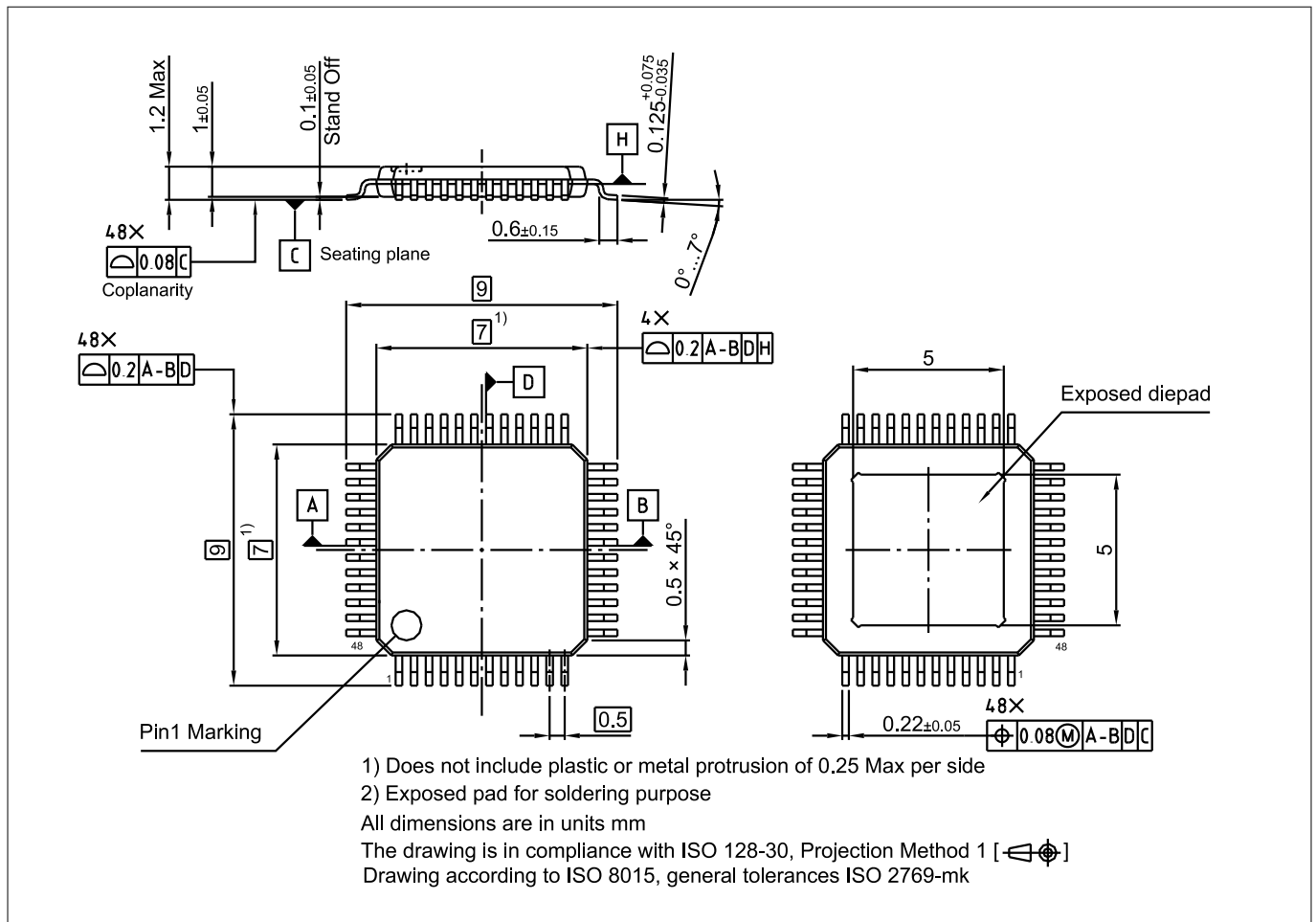


Figure 9 PG-TQFP-48

Green Product (RoHS compliant)

To meet the world-wide customer requirements for environmentally friendly products and to be compliant with government regulations, the device is available as a Green Product. Green Products are RoHS compliant (Pb-free finish on leads and suitable for Pb-free soldering according to IPC/JEDEC J-STD-020).

Information on packages

For more information on packages, such as recommendations on assembly, refer to www.infineon.com/packages.

10 Revision history

10 Revision history

Revision	Date	Changes
2.1	2026-03-16	Improvement of communication speed to 3Mbits, editorial updates and corrections
2.0	2023-11-02	Editorial updates and corrections
1.0	2022-01-24	Initial release of datasheet

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